Fiscal Year:	FY 2015	**Task Last Updated:**	FY 12/19/2014
PI Name:	Burma, Sandeep Ph.D.		
Project Title:	Mouse Glioma Models to Estimate Cancer Risks from HZE Particle Exposure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	78229	**Congressional District:**	21
Comments:	NOTE: Formerly at University of Texas Southwestern Medical Center at Dallas until fall 2019.		
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	03/11/2013	**End Date:**	09/10/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/10/2016 per NSSC information (Ed., 1/6/16)		
Key Personnel Changes/Previous PI:	December 2014 report: No key personnel changes.		
COI Name (Institution):	Bachoo, Robert M.D., Ph.D. (University of Texas Southwestern Medical Center at Dallas)		
Grant/Contract No.:	NNX13AI13G		
Performance Goal No.:			
Performance Goal Text:			
	There is a paucity of systematic studies carried out with well-defined model systems to evaluate cancer risks to astronauts from exposure to charged particles. Glioblastomas (GBM) are deadly brain tumors with very poor prognosis and are the third leading cause of cancer-related deaths among adults aged 30-50 years. Exposure to ionizing radiation is the only known risk factor for the development of these tumors. Unprecedented advances have recently been made in the genetic analysis of these tumors as part of the Human Cancer Genome Atlas Project. Five key genetic changes predominate in GBMs - loss of Ink4a, Arf, p53, and/or PTEN, and amplification of EGFR. Under the auspices of our currently-funded NASA grant, our lab has validated sensitive mouse models with brain-targeted deletions of these genes representing progressive steps in gliomagenesis. By using genetically “pre-sensitized” models, we are capable of rapidly assessing brain tumor development after exposure to particle radiation. Our current research shows that Fe ions are		

Task Description:	4-fold more gliomagenic than X-rays. Additionally, we have demonstrated that Fe-induced gliomagenesis is triggered by specific oncogenic events, such as MET amplification or PDGFR over-expression, that are also relevant to human GBMs. We are currently using further finessed versions of these mouse models to analyze not only the tumorigenic potential of an array of particles and doses, but also to identify the cell types in the brain that are most susceptible to transformation following charged particle exposure. Furthermore, we propose to undertake large-scale oncogenomic studies to identify pathways that are altered in particle-induced GBMs. The results of this study will greatly facilitate the estimation of cancer risks from charged particle exposure and should ultimately lead to radioprotective or therapeutic strategies to inhibit or treat these recalcitrant tumors. Specific aims are: Aim 1: Evaluation of the tumorigenic potential of HZE particles using Nestin-Cre driven Ink4a/b-/-Arff/f and p53+/f Pten+/f mouse GBM models. Aim 2: Identification of the “cell-of-origin” of HZE-induced gliomas using cell-type specific Cre-ERT2 mouse GBM models. Aim 3: Elucidating mechanism(s) underlying HZE induced gliomagenesis using an integrated oncogenomic appoach.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As cancer therapy with particles (protons and C ions) is becoming more widely used, our research with heavy ions is also important from a therapeutic standpoint here on Earth.
Task Progress:	In experiments conducted thus far we further characterized the Nestin-Cre Ink4a/b-/-Arff/f model that we are using to study radiation-induced gliomagenesis and we characterized a second, complementary, model system (Nestin-Cre p53+/f Pten+/f). As p53 mutations and Ink4a/Arf loss occur in a somewhat mutually-exclusive manner in human GBM, these mice allow us to address the question of radiogenic glioma using two independent models. Following exposure to 0.5 Gy of Fe ions, both models develop high grade glial tumors that phenocopy the histopathological and molecular features of human GBM, with a frequency of at least 25%. The key oncogenic changes occurring in these mice are the same as those seen in patients, namely loss of additional tumor suppressor genes and gain or activation of receptor tyrosine kinases (Met and Pdgfr). These changes led to the over-expression of the transcription factor Sox2 which appears to be essential for transformation and maintenance of a stem-cell phenotype within the tumor. Ex-vivo experiments further demonstrated the importance of Sox2 in maintaining a cancer stem cell phenotype in tumor cultures. In additional studies we sought to determine the transforming potential of an array of charged particles with different LETs (proton, carbon, silicon, and iron) in order to improve our understanding of the risk from GCR exposure. Here, we used the Nestin-Cre Ink4a/b-/-Arff/f mouse and demonstrated gliomagenesis following irradiation with all tested particles. Mice exposed to lighter ions (0.5 Gy of protons or carbon) formed tumors 5-10% of the time. This frequency increased up to 37.5% following exposure to silicon but then decreased with further increases in Z and LET. Ongoing studies with the Nestin-Cre p53+/f Pten+/f model tentatively corroborate results obtained with the Nestin-Cre Ink4a/b-/-Arff/f model. These results are suggestive of a linear relationship between LET of charged particles and GBM development which peaks at around an LET of 79 KeV/µm (silicon).
Bibliography Type:	Description: (Last Updated: 12/09/2025)
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